Documentation!

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Researched demand forecasting

Demand forecasting uses historical data and market trends to predict future product demand, helping organizations maintain optimal inventory levels and avoid shortages or excess stock. Demand fluctuations can often be explained by underlying factors such as seasonality, promotions, and consumer behavior.

Ultimate Goal:

Snowflake → Python ML (forecast engine) → Snowflake → Power BI

Where:

- Snowflake stores data.
- Claude is the reasoning layer.
- MCP gives Claude access to Snowflake and tools.
- Python contains forecasting functions/models.
- Power BI displays outputs.

AI agent can then:

1. Query Snowflake.
2. Call forecasting functions.
3. Compare forecasts to actuals.
4. Explain anomalies.
5. Write results back.

Reviewed Sample Historical Records for 2 products, converted to csv file

[sales]

A1 | fx Customer

	A	B	C	D	E
1	Customer	Organization	Item	Month	History
434	AMAZON.COM	PRD:IND:IND-GEODIS WHSE IND US	PAC00-14231	1-May-23	36165
435	AMAZON.COM	PRD:IND:IND-GEODIS WHSE IND US	PAC00-14231	1-Jun-23	37695
436	AMAZON.COM	PRD:IND:IND-GEODIS WHSE IND US	PAC00-14231	1-Jul-23	26860
437	AMAZON.COM	PRD:IND:IND-GEODIS WHSE IND US	PAC00-14231	1-Aug-23	16619
438	AMAZON.COM	PRD:IND:IND-GEODIS WHSE IND US	PAC00-14231	1-Sep-23	33163
439	AMAZON.COM	PRD:IND:IND-GEODIS WHSE IND US	PAC00-14231	1-Oct-23	32505
440	AMAZON.COM	PRD:IND:IND-GEODIS WHSE IND US	PAC00-14231	1-Nov-23	27516
441	AMAZON.COM	PRD:IND:IND-GEODIS WHSE IND US	PAC00-14231	1-Dec-23	20286
442	AMAZON.COM	PRD:IND:IND-GEODIS WHSE IND US	PAC00-14231	1-Jan-24	31118
443	AMAZON.COM	PRD:IND:IND-GEODIS WHSE IND US	PAC00-14231	1-Feb-24	27145
444	AMAZON.COM	PRD:IND:IND-GEODIS WHSE IND US	PAC00-14231	1-Mar-24	28719
445	AMAZON.COM	PRD:IND:IND-GEODIS WHSE IND US	PAC00-14231	1-Apr-24	35673
446	AMAZON.COM	PRD:IND:IND-GEODIS WHSE IND US	PAC00-14231	1-May-24	28244
447	AMAZON.COM	PRD:IND:IND-GEODIS WHSE IND US	PAC00-14231	1-Jun-24	24961
448	AMAZON.COM	PRD:IND:IND-GEODIS WHSE IND US	PAC00-14231	1-Jul-24	35881
449	AMAZON.COM	PRD:IND:IND-GEODIS WHSE IND US	PAC00-14231	1-Aug-24	30348
450	AMAZON.COM	PRD:IND:IND-GEODIS WHSE IND US	PAC00-14231	1-Sep-24	30034
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

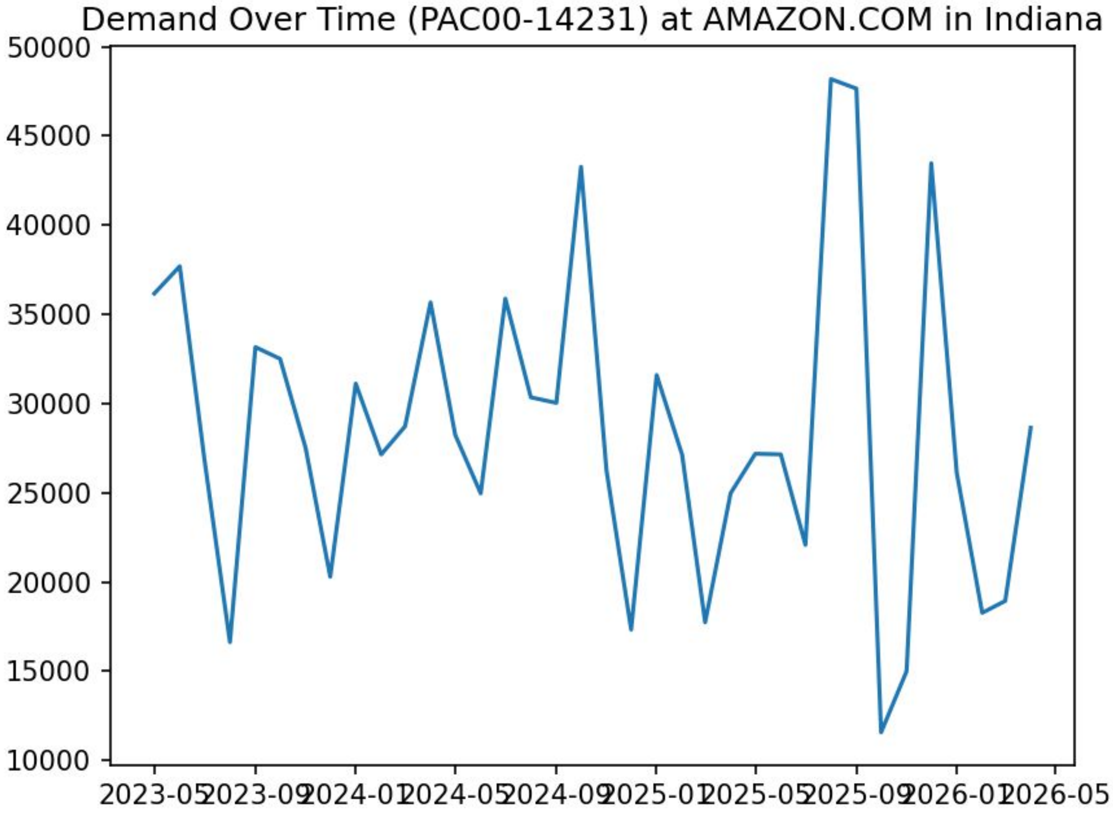
Basic python functions to gather data about the database

```python
import pandas as pd # Reads/manages data tables.
import matplotlib.pyplot as plt # Makes graphs
from statsmodels.tsa.holtwinters import ExponentialSmoothing # Basic forecasting

df = pd.read_csv("sales_data.csv")

print(df.head()) # (prints first few lines)
df.info()
print(df.isna().sum())

print(df["Customer"].unique()) # prints all our customers
```

Isolated 1 customer, 1 warehouse, & 1 product

```python
# filter data
df_one = df[
    (df["Customer"] == "AMAZON.COM") &
    (df["Item"] == "PAC00-14231") &
    (df["Organization"] == "PRD:IND:IND-GEODIS WHSE IND US")
].copy()
df_one["Month"] = pd.to_datetime( # converting dates
    df_one["Month"],
    format="%d-%b-%y"
)
df_one = df_one.dropna(subset=["History"]) # remove missing rows
df_one = df_one.groupby("Month", as_index=False)["History"].sum() # aggregate monthly demand
df_one = df_one.set_index("Month") # add monthly frequency
df_one = df_one.asfreq("MS")
# plot the demand
plt.plot(df_one.index, df_one.values) # draw line graph
plt.title("Demand Over Time (PAC00-14231) at AMAZON.COM in Indiana")
plt.show() # display our graph
# forecast model
model = ExponentialSmoothing(
    df_one["History"],
    trend="add"
)
fit = model.fit() # learn past demand pattern

forecast = fit.forecast(18)  # predict next 18 months
print("forecast for next 18 months:")
print(forecast)
```

Plotted diagram and made 18-month moving avg forecast




```
forecast for next 18 months:
2026-05-01    24997.913754
2026-06-01    24777.775965
2026-07-01    24557.638175
2026-08-01    24337.500386
2026-09-01    24117.362597
2026-10-01    23897.224808
2026-11-01    23677.087019
2026-12-01    23456.949229
2027-01-01    23236.811440
2027-02-01    23016.673651
2027-03-01    22796.535862
2027-04-01    22576.398073
2027-05-01    22356.260284
2027-06-01    22136.122494
2027-07-01    21915.984705
2027-08-01    21695.846916
2027-09-01    21475.709127
2027-10-01    21255.571338
```

Researched forecasting w Intro to Materials Management

Forecasting is estimating future demand to plan resources before orders arrive — it's inevitable for any firm. Demand varies due to trend, seasonality, and random variation, and only independent demand items need forecasting (dependent demand is calculated). Key intrinsic methods are moving averages (simple, lags trends) and exponential smoothing (weighted, more responsive). Forecast accuracy is measured via MAD and tracked with a tracking signal to catch bias early.

file:///C:/Users/prsha/5-26%20forecast/Introduction%20to%20Materials%20Management.pdf

Snowflake

AI Overview

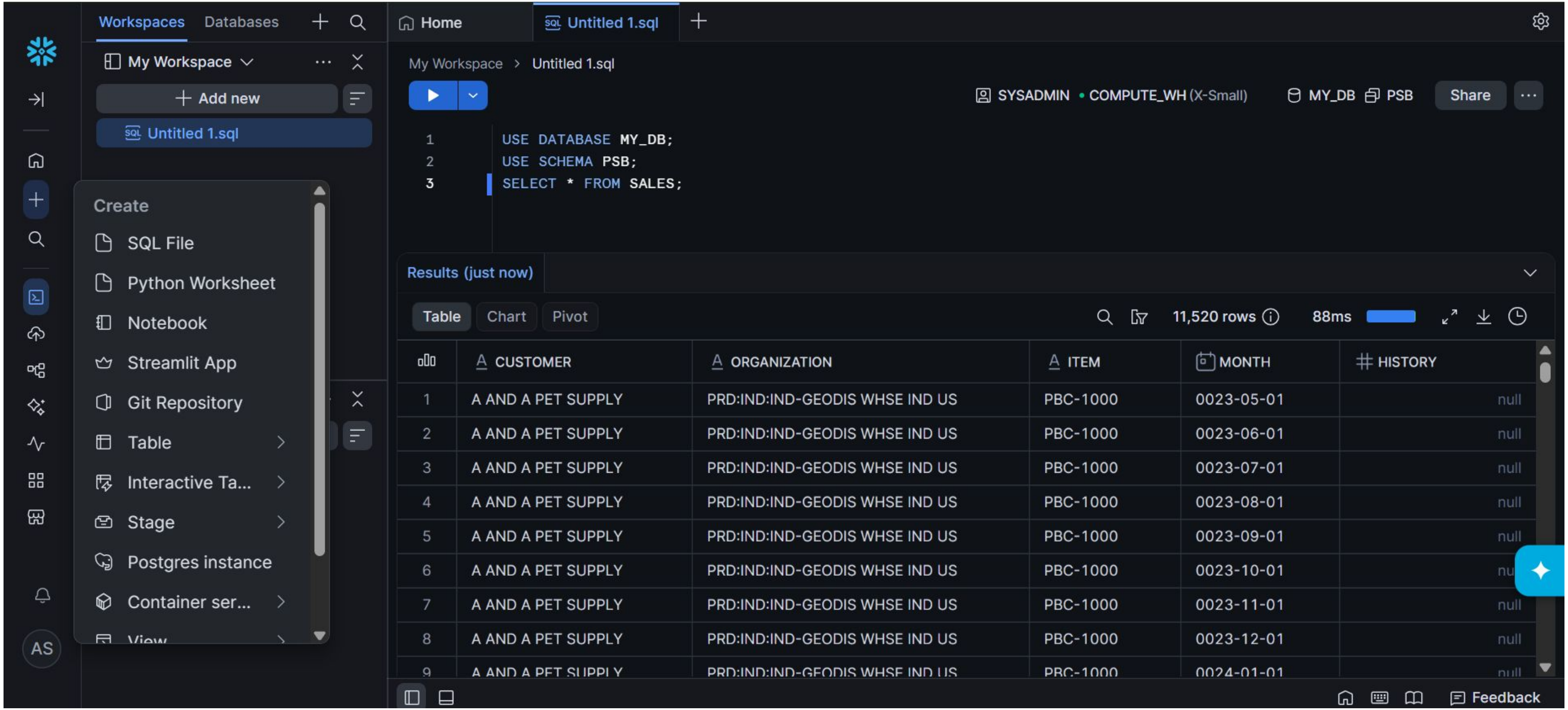
Snowflake is a fully managed, cloud-based data and AI platform that allows organizations to store, analyze, and share massive amounts of data without needing to manage complex physical infrastructure. Reddit · r/snowflake +2

Core Capabilities

- **Data Warehousing & Data Lakes:** It excels at consolidating and querying structured and semi-structured data (like JSON) at lightning speed. SnapLogic +3
- **AI & Machine Learning:** The platform natively supports running AI/ML models, building data agents, and processing unstructured data securely. Reddit · r/snowflake +1
- **App Development & Sharing:** Developers can build data-intensive applications and securely share or monetize live data with external partners through the Snowflake Marketplace.

Built a database, schema, and table here

AS seen in SQL:



MCP Connectors

Researching to Connect external MCP (Model Context Protocol) servers to use with Cortex Code in Snowsight and Snowflake Intelligence agents.

Snowflake
↓
Historical demand data

Python
↓
Groups data by SKU

Claude API
↓
Chooses forecasting method

Python
↓
Calculates forecast

Power BI
↓
Dashboard